YUHESEN

FR-mini Ackermann Steering

Drive-by-wire Chassis

User manual V1.0.3

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1. Foreword

(1) Thank you for purchasing our product, this user manual is applicable to FR-mini Ackerman Drive-by-wire Chassis (hereby referred to as "FR-mini").

(2) Before use, please carefully read this user manual and attentions, and correctly use strictly in accordance with this manual.

(3) For the loses caused by serious violation of this user manual, we undertake no responsibilities.

(4) Please well keep this manual for user reference during your operation.

(5) Professionals are required for commissioning, connection and installation of the chassis equipment to avoid irretrievable loses.

(6) DO NOT install, remove or replace equipment lines with electricity. If it is necessary to commission this product with electricity, please select the special commissioning tools with good insulation.

(7) Please use this product under the conditions allowed by laws and regulations, so that the public property or life safety will not be affected.

(8) We will irregularly update this product, the contents of update will be added into the new manual without notification.

(9) This manual may contain the contents which are not correct in technology or which do not comply with the operation. In case of problems which cannot be solved during use of this manual, please contact with the customer service or technical department of us.

(10) As for the contents of this manual, we will try our best to ensure that they are correct and accurate. In case of any improper or incorrect contents, please contact us for confirmation, thank you!

Safety Information

The information herein does not include how to design, install or operate a complete robot, nor the peripheral equipment which may affect the safety of this complete system. The design and use of the complete system comply with the safety requirements formulated in the national standards and specifications. The integrators and end customers of FR-mid are responsible for being sure to comply with practical laws and regulations of relevant countries to ensure that the application of the complete robot will not cause any major danger. These include but are not limited to the following:

■ Effectiveness and responsibilities:

- A risk evaluation shall be conducted to the complete robot system. All the additional safety equipment of other machineries defined by risk evaluation shall be connected. It shall be ensured that, the design and installation of the peripheral equipment of the whole robot system, including software and hardware system, are correct.
- This robot is not equipped with relevant safety functions that a complete autonomously moveable robot shall have, including but not limited to automatic collision avoidance, fall prevention and alarm for creature approaching, etc. For relevant functions, the integrators and end customers are required to conduct safety evaluation in accordance with relevant regulations and feasible laws and regulations to ensure that the developed robot has no any major danger or potential safety hazard during actual application.
- Collecting all the documents of technical files: Including risk evaluation and this manual. Before operation and use of equipment, the existing safety risks may be known.

■ Environments:

- For first use, please carefully read this manual to understand the basic contents and operation specifications.
- For remote operation, please select the areas which are relevantly open. This chassis is not equipped with any sensor for automatic obstacle avoidance.

- This chassis shall be used under the temperature of -20°C~50°C.
- The chassis is not customized for IP protection grade, the IP protection grade of this chassis is IP22.

Inspection:

- Inspecting to ensure that the batteries of the equipment are full. Ensuring that the chassis has no abnormality. Inspecting whether the battery of the remote controller is full.

Operation:

- Please make sure that the remote control is on when you use it for commissioning, and make sure that the vehicle can receive the remote control commands.
- Ensuring that operation is conducted in a relatively open place. And remote control shall be conducted with sight distance.
- FR-mini The maximum load is 20KG, during use, it shall be ensured that the effective load does not exceed 20KG.
- In case of alarm of low battery of the equipment, please charge timely. In case of equipment abnormality, please stop use immediately to avoid secondary damage.
- In case of equipment abnormality, please contact relevant technicians, DO NOT process without permission.
- Please use the equipment in the environment which meets the IP protection grade requirements of the equipment.
- DO NOT directly push the chassis.
- During charging, please ensure that the environment temperature is higher than 0°C.

Maintenance:

- In case of serious tire wearing, please replace timely.
- If the battery will not be used for a long time, when the battery is fully charged, please charge the battery regularly in each month.
- The battery shall be charged once a month at least.

2. Introduction

FR-mini is a versatile drive-by-wire robotics mobile platform, it adopts Ackerman front steering, and rear drive form. Compared with the chassis of differential drive form on the ordinary pavement, FR-mini has a faster traveling ability and relatively strong load capacity. At the same time, the wearing of tire is lighter, matching with whole bridge suspension, the chassis can pass through the common obstacles, such as speed bump, etc. Therefore, it is more applicable for long-term outdoor traveling; And this chassis is a underlayer control system structure based on VCU vehicles control, it uses CAN bus management, having the features of high precision and modularization, etc. By the modules and navigation systems of LiDAR, GPS and manipulator, etc., this chassis is widely used in autonomous driving, unmanned patrol, logistics, transportation distribution, scientific research and various new applications and explorations requiring for mobile chassis.

2.1. Product List

After cargo is delivered, please carefully confirm the product list：

Chassis *1

Remote controller *1

Charger (24V) *1

Product manual *1

2.2. Performance Parameters

Table 2-1 FR-mini Parameters

Parameter Type	Performance	Parameter
Structural size and weight	Dimensions(W*D*L)	615*405*190mm
	Weight	18kg
	Drive	Front Ackermann steering, rear motor drive
	Suspension	Independent suspension
	Material	Q235
	Ground clearance	61mm
	Wheelbase	400mm
	Wheel track	312mm
	Tire type/diameter	140mm
Basic configuration	Driving motor	100W*2 servo motor
	Steering motor	65W servo motor
	Battery type	24V/10AH lithium battery/BMS 24V/20AH lithium battery/BMS(optional)
	Charging time	2-3h
	Charging method	24V/10A manual charging adapter
	External power supply	24V/15A-12V/15A
	Braking mode	Motor brake
	Parking method	Motor parking
	Turn signal light	-
	Brake lamp/deceleration indicator/fault indicator	-
	Wheel speed sensor	-
Safety measures	Emergency stop button	√
	Front and rear bumper strip	-
	Command verification	√
	Heartbeat protection	√
	Fault handling for steering system	√

	Fault handling for braking system	√
	Fault handling for driving system	√
	Emergency power off parking protection	√
	Battery fault monitoring and protection	√
	Online detection for whole vehicle CAN node	√
	Whole vehicle fault level division and processing	√
	Vehicle fault warning	√
	Prompt of fast vehicle deceleration	√
	Processing of remote controller disconnection	√
	Charging safety monitoring and protection	√
VCU configuration	Main frequency	168MHz
	Hardware floating point acceleration	√
	Movement control	√
	Communication interface	CAN Interface
	Communication protocol	CAN 2.0B
Performance parameters	Remote control distance	100m
	Vertical load (level road)	20kg
	Speed	6.5km/h
	Mileage	24V10AH - 17km(full loads)/40km(no load) 24V20AH - 35km(full loads)/80km(no load)
	Minimum turning radius	1.1m
	Wading depth	30mm
	Maximum climbing angle	10°(full loads)
	Span width	90mm(full loads)
	Obstacle surmounting height	30mm(full loads)
	Steering angle	≤0.5°
	IP rating	IP22
	Operating temperature	-20℃~50℃

	Storage temperature	0℃~40℃

3. Product presentation

The contents in this part are only the basic introductions for FR-mini Ackermann Drive-by-wire Chassis, facilitating the users and developers to know FR-mini chassis basically. As shown in Figure 3-1 and Figure 3-2, there are the front and rear overall figure of the whole Ackermann drive-by-wire chassis.

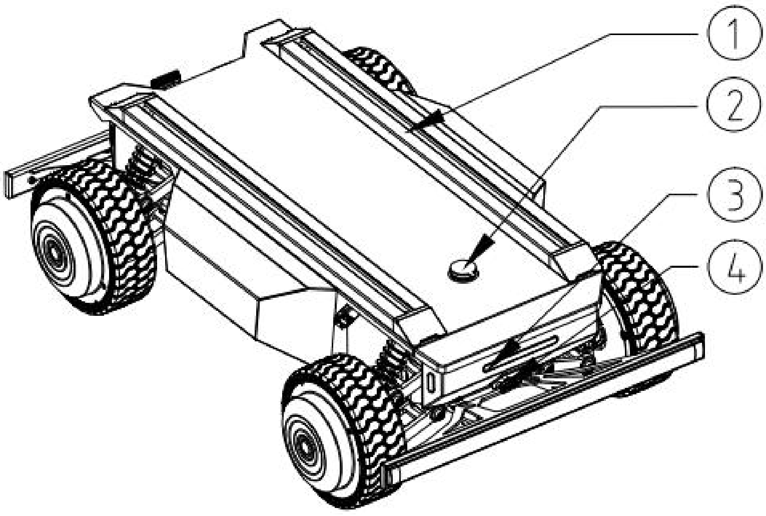


Figure 3-1rear view

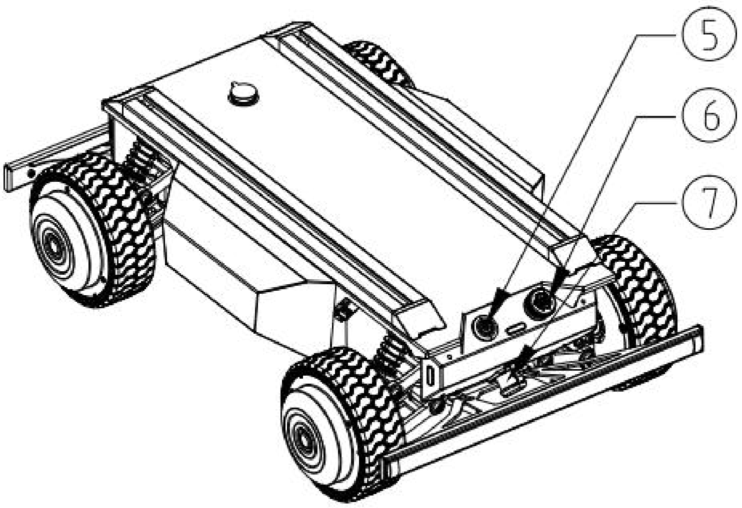


Figure 3-2 front view

Note:
① 1640Q mounting profile;
② Top electrical interface;
③ Front light;
④ Debugging/firmware upgrade port;
⑤ Power switch;
⑥ Emergency stop button;
⑦ Charging port;

The FR-mini adopts a modular design approach overall, featuring high safety and reliability. Structurally, it employs a front Ackermann steering system and a rear solid axle suspension with a body-on-frame design, ensuring excellent body strength and rigidity, which enhances overall vehicle safety. It boasts strong impact resistance and anti-jolting performance, along with superior off-road capability, enabling it to navigate complex road conditions.

The braking system utilizes an electric motor design. Additionally, an emergency stop switch is installed at the rear of the vehicle. When pressed in an emergency, it immediately halts the vehicle, achieving full control. The emergency stop switch also supports functional detection—if it is damaged or disconnected, the VCU prevents the vehicle's drive system from powering up. Multiple safeguards ensure safe vehicle operation.

The chassis features integrated control, with the VCU performing unified analysis and judgment of vehicle signals to form a closed-loop control system. It enables fault diagnosis and timely safety protection measures, reliably achieving remote unmanned vehicle monitoring. The vehicle roof is equipped with open 24V and 12V electrical interfaces, as well as communication ports, and comes with standard mounting brackets for rapid secondary development by users.

3.1. State Indicator

Users can determine the vehicle's status through the battery level display and startup sound on the remote control. For details, please refer to Table 3-1.

Status	Description
Battery remaining capacity	The current battery SOC can be viewed by swiping left on the remote control display screen and checking the Value in Ext.V (Figure 3-3). As shown in Figure 3-3, the current battery SOC is 48.62%.

Fault indicator	The current vehicle faults can be obtained through CAN signal feedback, including the minor fault signals and fault levels: Level 1 fault: CAN signal alarm; Level 2 fault: CAN signal alarm and vehicle speed reduction; Level 3 fault: CAN signal alarm and vehicle stop.

Table 3-1 Chassis status description

Name	ID	Value
TX. V	0	5.55V
Int. V	0	4.96V
Sig. S	0	10
Ext. V	1	48.62V

Figure 3-3 Remote controller vehicle voltage monitoring interface

Note: This interface appears by swiping left on the remote control screen.

- **TX.V**: Current battery voltage of the remote control
- **Int.V**: Power supply voltage of the receiver
- **Sig.S**: Signal strength of the receiver
- **Ext.V**: Battery state of charge (SOC)
- **ID 0**: Signal from the remote control transmitter or receiver
- **ID 1**: First sensor connected to the receiver, and so on.

3.2. Instructions of electrical interface

3.2.1. Instructions of top electrical interface

The FR-mini is equipped with a M19-15 core aviation electrical connector at the top. This electrical interface is configured with three different power supply groups and one CAN communication interface, with pre-wired leads for easy connection. This allows users to provide power and communication support for various expansion devices.

For the specific location, please refer to the **Top Electrical Interface Layout Diagram (Figure 3-4)**.

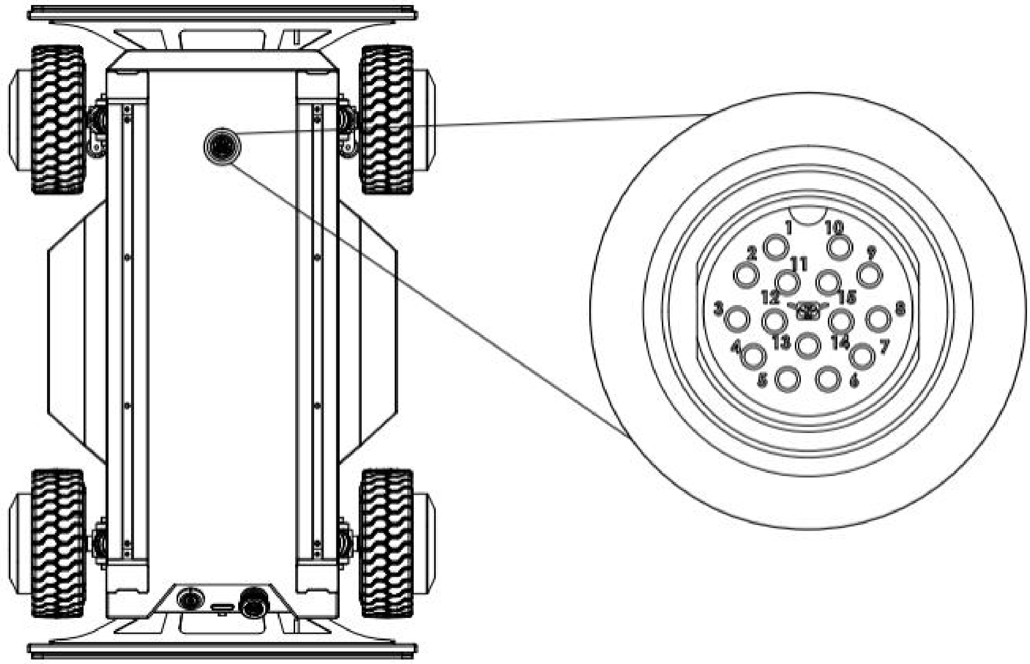


Figure 3-4 Top Electrical Interface Layout Diagram

The specific pin definitions of top electrical interfaces are shown in Table 3-3 below.

Pin	Type	Definitions	Remark
1/2	Power output	12V+	12V
9/10/15		12V-	
5/6		BAT_24V+	Battery output
7/8		BAT_24V-	
13	Communication	CAN2_H	CAN communication
14		CAN2_L	

Table 3-2 Definitions of top electrical interfaces

Please note that the extended power supply here is internally controlled. When the battery voltage is too low, the BMS (Battery Management System) will activate protection and stop battery discharge. Users should pay attention to charging during operation.

3.2.2. Instructions of electrical interface

The electrical panel at the chassis is shown in **Figure 3-5**, where:

- **B1**: debugging/firmware upgrade port
- **B2**: charging port

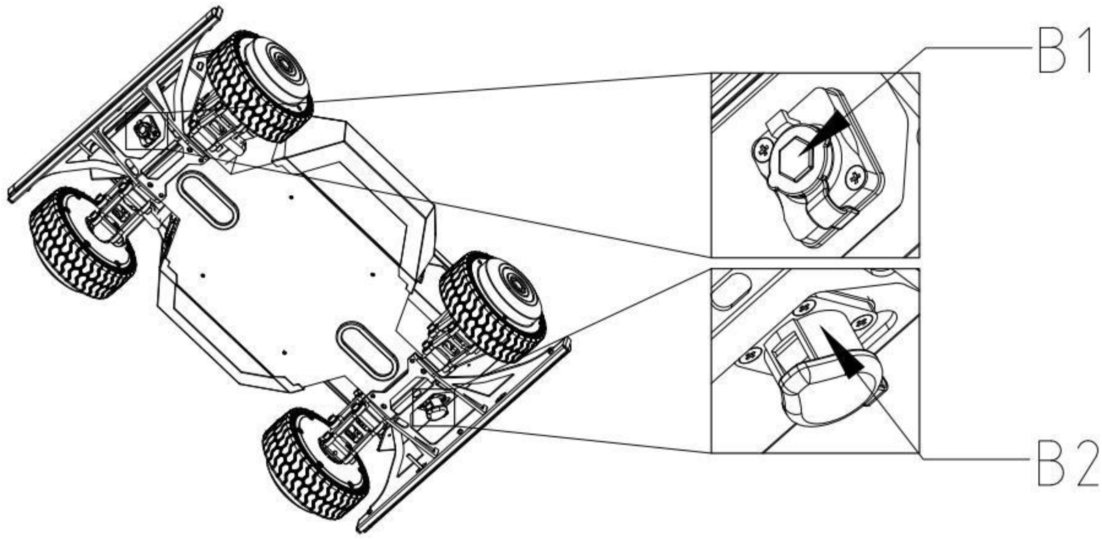


Figure 3-5 Rear electrical panel

3.3. FS-i6S Remote control instructions

The remote controllers have been successfully paired before leaving the factory and require no further adjustments.

Altering the remote controller settings without authorization may cause control confusion, loss of control, and other issues. Please do not modify the settings without proper reason.

If you encounter any parameter configuration problems, please contact our customer service or technical support team.

Should modifications be necessary, they must be performed by qualified technical personnel only.

3.3.1. FS-i6S control instruction

Each FR-mini unit is equipped with an FS-i6S remote controller, allowing users to easily operate the FR-mini.

In this product, the FS-i6S remote controller is configured with the following control scheme:

- **Left hand**: N/A
- **Right hand**: Forward/backward throttle and left/right steering

For detailed definitions and functions, please refer to **Figure 3-6**.

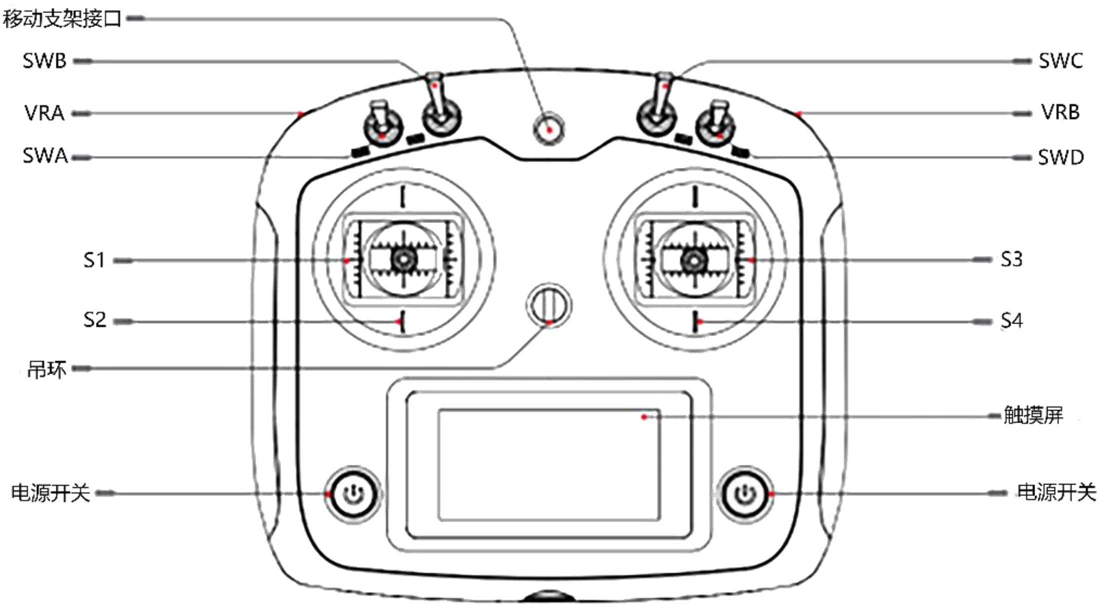


Figure 3-6 FS-i6S Schematic diagram of the FS-i6S remote controller's operation panel

(1) SWA is the control mode toggle switch with two positions. With the remote controller facing upward:

- **Upper position**: Remote control mode
- **Lower position**: Command control mode

(2) SWB is the gear shift toggle switch with three positions:

- **Middle position (N)**: Vehicle is in neutral and does not respond to forward/backward motion signals
- **Upper position (D)**: Chassis receives forward motion signals from the **S4** joystick to move forward
- **Lower position (R)**: Chassis receives backward motion signals from the **S4** joystick to move backward

(3) VRB is the parking request control knob:

- **Upward turn**: Sends a parking request to engage the parking brake
- **Downward turn**: Sends a release request to disengage the parking brake

The VRB serves as the enable switch for the remote's speed joystick (S4)—when pulled and held down, speed control follows the S4 joystick's input.

(4) S2 is not available:

(5) S4 is the throttle joystick, controlling the **FR-mini**'s forward/backward speed. **S3** controls front-wheel steering.

(6) SWC adjusts the **S4** joystick's speed mode (using forward gear as an example):

- **Top position**: Low-speed mode
- **Middle position**: Medium-speed mode
- **Bottom position**: High-speed mode

(7) Power switch: Controls the remote's power state:

- **To turn on**: Long-press both power switches on the display sides when powered off
- **To turn off**: Long-press both switches when powered on
- **If the receiver is active**, the remote cannot be turned off via the switches—remove the battery instead.

3.3.2. Remote controller alarm instruction

Switch Position Alarm	If the toggle switches (SWA/SWB/SWC/SWD) are not in their default positions during startup, an alarm screen will appear, prompting the user to move all switches to the upward position. The system will proceed to the main interface only when all switches are in their default positions.
Low Voltage Alarm	When the voltage drops below the alarm threshold, the system triggers an alarm, causing the remote controller's screen to flash: **TX icon flashes** if the remote's voltage is too low. **RX icon flashes** if the chassis voltage is too low.
Communication Failure Alarm	If the control distance between the remote and chassis is too far or obstructed by environmental interference, the signal strength may drop. A **signal strength below 5** triggers a communication failure alarm, warning the user of weak signal conditions.
Remote Controller Inactivity Alarm	If the remote remains unused for an extended period, its buzzer will emit intermittent alarms.
Shutdown Alarm	During shutdown, the remote checks whether the chassis is powered off. If the chassis is still on, a warning screen appears, requiring the chassis to be powered off before the remote can shut down. (To force shutdown the remote while the chassis is still on, remove the remote's battery.)

Table 3-3 Remote controller alarm status description

3.3.3. Control command and movement instruction

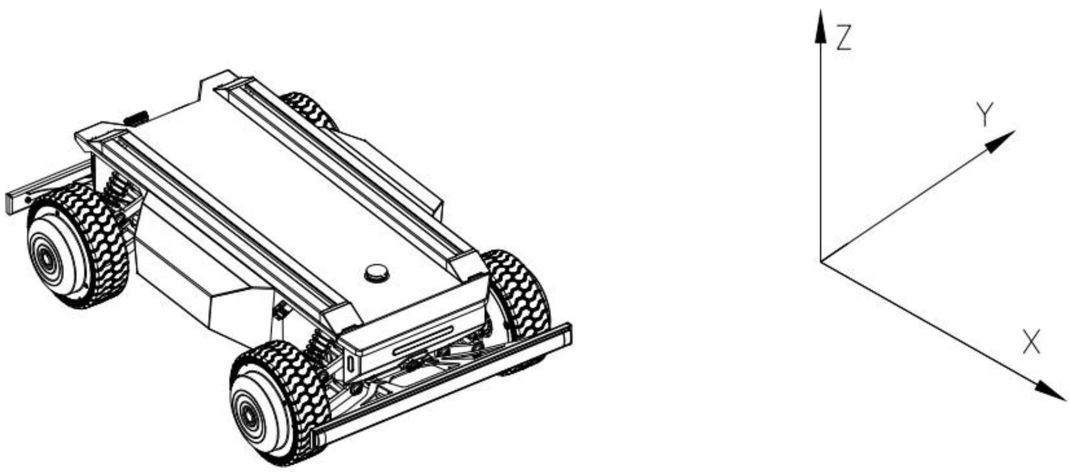


Figure 3-7 Chassis coordinate reference frame

We establish the coordinate reference frame for ground mobile vehicles in accordance with ISO 8855, as shown in Figure 3-7.

As illustrated in Figure 3-7, the FR-mini vehicle body is aligned parallel to the X-axis of the defined reference coordinate system.

Remote Control Mode:

Movement Control:

- Pull down the **VRB knob** to disengage the parking gear.
- Set **SWB toggle** to **D (Drive) gear**—pushing the **S4 joystick forward** moves the vehicle in the **+X direction**.
- Set **SWB toggle** to **R (Reverse) gear**—pulling the **S4 joystick backward** moves the vehicle in the **–X direction**.
- When the **S4 joystick is pushed fully forward/backward**, the speed in the **±X direction** depends on **SWC's high/medium/low speed setting**.

Steering Control:

- The **S3 joystick** controls the front-wheel steering:
- **Push left** → **left turn** (maximum left steering angle at full deflection).
- **Push right** → **right turn** (maximum right steering angle at full deflection).

Command Control Mode:

- **Target gear command values:**
- **04**: Movement in the **+X direction**.
- **02**: Movement in the **–X direction**.

3.3.4. Steering angle control and feedback instruction

Both the vehicle's steering angle control and feedback correspond to the **inner wheel steering angle** during operation:

- **Left turn** = **Left wheel steering angle** (positive value)
- **Right turn** = **Right wheel steering angle** (negative value)

As shown in **Figure 3-9 (Ackermann vehicle model)**, the remote controller and CAN control/feedback commands all refer to the **α angle**.
The equivalent bicycle model steering angle is denoted as **β**.

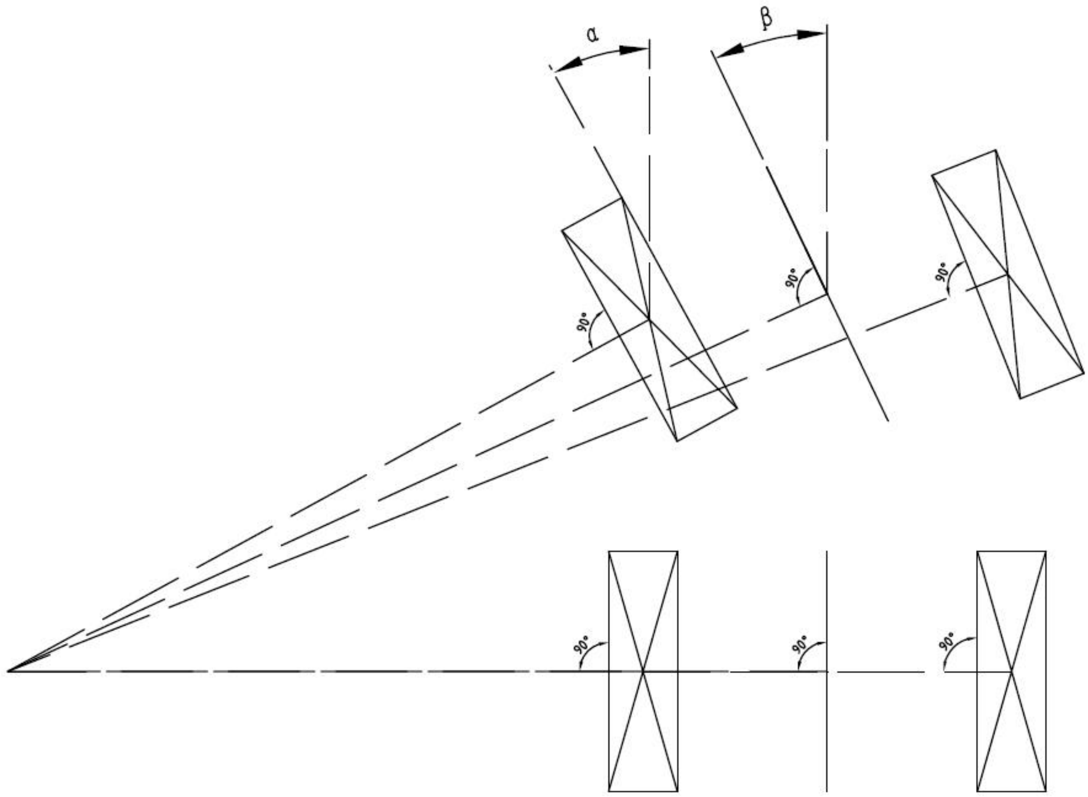


Figure 3-9 Ackermann vehicle model

4. Getting Started

This section primarily introduces the basic operations and usage of the FR-mini platform, detailing how to perform secondary development of the vehicle body through the CAN bus protocol.

4.1. Use and operation

Basic Remote Control Operation Procedure

Inspection

1. **Check the vehicle status.**

- Inspect for any visible abnormalities. If found, contact after-sales support.

2. **Verify the emergency stop switch status.**

- Ensure all rear emergency stop switches are in the **released position**.

3. **Confirm all remote control gear positions are in neutral.**

Startup

1. **Power on the remote controller.**

- Press and hold the power buttons on both sides of the display.

2. **Press B1 (Power Switch).**
3. **Check the remaining battery level on the remote.**

- If below **30%**, recharge before operation.

4. **Release brakes and parking lock, then switch to remote driving mode.**

- Observe if the brake lights are flashing, indicating a fault.
- If a fault is detected, connect a **CAN card** to read error codes and signals, then contact after-sales support for resolution.

Shutdown

- Press **B1 (Power Switch) again**, then release to power off.

Emergency Stop

- Press the **emergency stop switch** on the FR-mini's rear electrical panel.
- The vehicle will enter **emergency stop mode**, with **left & right turn signals flashing** as an alarm.

4.2. Charge

The FR-mini mobile robot chassis comes standard with a 24**V/5A charger** to meet users' charging needs.

Charging Procedure:

1. **Before charging**, ensure the FR-mini is **powered off** and confirm that **B1 (Start Switch)** on the rear electrical panel is in the **OFF position**.

2. **Connect the charger:**

- First, plug the charger’s **output connector** into **charging port** on the rear of chassis.
- Then, insert the charger’s **AC plug** into a **220V AC power outlet**.

3. **After charging:**

- First, **unplug the AC plug**, then **remove the output connector**.

4. **Charger status indicators** are described in **Table 4-1**.

LED indicator status	Charger status
LED1 in bright red	The charger's input cable connector is already energized.
LED2 in bright red	incidate charger is in charging
LED2 in bright green	indicate battery is fully charged

Table 4-1 Charger status indicators description

If the charging environment temperature is too high, the charger may enter temperature protection mode. Please move the charger to a cool or well-ventilated area. Normal charging will resume when the internal temperature drops below 60°C. The charger protection status descriptions are shown in Table 4-2.

Protection Features	**Description**
Overheat Protection	The charger automatically stops charging when the internal temperature reaches the overheat protection threshold.
Output Short-Circuit Protection	The charger automatically shuts off the output in case of an accidental short circuit.
Reverse Polarity	The charger disconnects the internal circuit from the battery if the

Protection	battery is connected in reverse.
Output Overvoltage Protection	The charger automatically shuts off the output if an overvoltage condition is detected.

Table 4-2 Charger Protection Status Description

Note:
During vehicle charging, the VCU (Vehicle Control Unit) performs integrated charging protection. If charging is initiated while the vehicle is powered on, the following safety measures are activated:

- The motor driver is controlled to shut down high-voltage power for safety.
- Normal operation automatically resumes after charging is disconnected.

Additionally:

- Corresponding CAN signal flags are transmitted to indicate the charging status.
- If necessary, a manual override command can be sent to terminate the charging shutdown process.

4.3. Development

The FR-mini product provides developers with a CAN interface for vehicle control commands, enabling direct communication with the vehicle's electronic systems

4.3.1. CAN protocol

The FR-mini product uses the CAN 2.0B extended frame for communication, with the message format in Intel format and a baud rate of 500K. Through the external CAN bus interface, it is possible to control the chassis movement, including vehicle speed, steering angle, brake pedal opening, and parking requests. The FR-mini will provide real-time feedback on the current motion status information as well as the system status information of the FR-mini chassis.

The specific protocol details are as follows:

The motion command control frame includes gear control, vehicle speed control, steering angle control, parking request, and checksum. The specific protocol content is shown in Table 4-3. For wiring instructions, please refer to Section 4.3.2. For CAN communication transmission requirements and test examples, please refer to Section 4.3.3.

Note: The CAN interface is a non-isolated interface. During use, ensure that the CAN wires are correctly connected and avoid connecting the CAN bus to power lines. Incorrect connections may damage the VCU.

CAN protocol as below:

Table 4-3 Command Control Frame and System Feedback Frame

Movement control command - control frame									
Message Name			**ID**				**Cycle (ms)**		**Message Length (Byte)**
auto_spd_ctrl_cmd			0x18C4D2D0				10		8
Signal Description	**Format**	**Starting Byte**	**Starting Bit**	**Signal Length**	**Data Type**	**Precision**	**Offset**	**Unit**	**Signal Value Description**
Target gear	Intel	0	0	4	Unsigned	1	0		00： disable 01： Gear P 02： Gear R 03： Gear N 04： Gear D
Target vehicle speed	Intel	0	4	16	Unsigned	0.001	0	m/s	0.001m/s/bit；
Target vehicle steering angle	Intel	2	20	16	signed	0.01	0	°	0.01°/bit；
Target vehicle braking	Intel	4	36	8	Unsigned	1	0		
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.

Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6
I/O control command - control frame									
Message Name			**ID**				**Cycle (ms)**		**Message Length (Byte)**
io_cmd			0x18C4D7D0				50		8
Signal Description	**Format**	**Starting Byte**	**Starting Bit**	**Signal Length**	**Data Type**	**Precision**	**Offset**	**Unit**	**Signal Value Description**
I/O control enabled(reserved)	Intel	0	0	1	Unsigned	1	0		0 = off 1 = on
Steering lamp switch(reserved)	Intel	1	10	2	Unsigned	1	0		0 = off 1 = left steering lamp on 2 = right steering lamp on
Clearance lamp switch(reserved)	Intel	1	13	1	Unsigned	1	0		0 = off 1 = on
Charging-forced power-on flag bit	Intel	5	40	1	Unsigned	1	0		Enabling this flag during charging forces the vehicle's 24V system to power on, allowing the vehicle to resume driving control. However, when this flag is enabled during charging, the vehicle cannot move in reverse.
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum

									value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6
Movement control command - feedback frame									
Message Name			**ID**				**Cycle (ms)**		**Message Length (Byte)**
ctrl_fb			0x18C4D2EF				10		8
Signal Description	**Format**	**Starting Byte**	**Starting Bit**	**Signal Length**	**Data Type**	**Precision**	**Offset**	**Unit**	**Signal Value Description**
Target gear feedback	Intel	0	0	4	Unsigned	1	0		00: disable 01: Gear P 02: Gear R 03: Gear N 04: Gear D
Current vehicle speed feedback	Intel	0	4	16	Unsigned	0.001	0	m/s	0.001m/s/bit;
Current vehicle steering angle feedback	Intel	2	20	16	signed	0.01	0	°	0.01°/bit;
Current vehicle braking status feedback	Intel	4	36	8	Unsigned	1	0		
Current vehicle operating mode feedback	Intel	5	44	4	Unsigned	1	0		0: remote control mode 2: auto command control mode 4: parking mode
Current acceleration mode (reserved)	Intel	6	48	2	Unsigned	1	0		00: normal 01: ECO mode 02: Sport mode Note: Reserved

									options will be enabled after torque adaptation is completed.
Current energy recovery mode (reserved)	Intel	6	50	2	Unsigned	1	0		00: Light energy recovery 01: Strong energy recovery Note: Reserved options will be enabled after torque adaptation is completed.
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6

Left rear wheel information feedback

Message Name	ID	Cycle (ms)	Message Length (Byte)
lr_wheel_fb	0x18C4D7EF	10	8

Signal Description	Format	Starting Byte	Starting	Signal	Data Type	Precision	Offse	Unit	Signal Value Description

			Bit	Length			t		
Current left rear wheel speed feedback	Intel	0	0	16	signed	0.001	0	m/s	0.001m/s/bit;
Current left rear wheel pulse count feedback	Intel	2	16	32	signed	1	0	1	4096 pulses per wheel revolution
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6
Right rear wheel information feedback									
Message Name			**ID**				**Cycle (ms)**		**Message Length (Byte)**
rr_wheel_fb			**0x18C4D8EF**				10		8
Signal Description	**Format**	**Starting Byte**	**Starting Bit**	**Signal Length**	**Data Type**	**Precision**	**Offset**	**Unit**	**Signal Value Description**
Current right rear wheel speed feedback	Intel	0	0	16	signed	0.001	0	m/s	0.001m/s/bit;
Current right rear wheel pulse count feedback	Intel	2	16	32	signed	1	0	1	4096 pulses per wheel revolution
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6
I/O control command - feedback frame									

Message Name			ID				Cycle (ms)		Message Length (Byte)
io_fb			**0x18C4DAEF**				50		8
Signal Description	**Format**	**Starting Byte**	**Starting Bit**	**Signal Length**	**Data Type**	**Precision**	**Offset**	**Unit**	**Signal Value Description**
I/O control enabling status feedback	Intel	0	0	1	Unsigned	1	0		0 = off 1 = on
Steering lamp switch status feedback	Intel	1	10	2	Unsigned	1	0		0 = off 1 = left steering lamp on 2 = left steering lamp on
Brake lamp switch status feedback	Intel	1	12	1	Unsigned	1	0		0 = off 1 = on
Clearance lamp switch status feedback	Intel	1	13	1	Unsigned	1	0		0 = off 1 = on
Charging-forced power-on flag bit	Intel	5	40	1	Unsigned	1	0		Enabling this flag during charging forces the vehicle's 24V system to power on, allowing the vehicle to resume driving control. However, when this flag is enabled during charging, the vehicle cannot move in reverse.
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6

Odonmeter feedback

Message Name			ID				Cycle (ms)		Message Length (Byte)
odo_fb			0x18C4DEEF				10		8
Signal Description	**Format**	**Starting Byte**	**Starting Bit**	**Signal Le**	**Data Type**	**Precision**	**Offset**	**Unit**	**Signal Value Description**

				ng th					
Total mileage	Intel	0	0	32	signed	0.001	0	m	0.001m/bit
Total angle(reserved)	Intel	4	32	32	signed	0.001	0	rad	0.001rad/bit

Battery BMS information feedback

Message Name	ID	Cycle (ms)	Message Length (Byte)
bms_Infor	0x18C4E1EF	100	8

Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Current battery voltage	Intel	0	0	16	Unsigned	0.01	0	V	0.01V/bit;
Current battery current	Intel	2	16	16	signed	0.01	0	A	0.01A/bit;
Current remaining battery capacity	Intel	4	32	16	Unsigned	0.01	0	Ah	0.01Ah/bit;
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6

Battery BMS flag bit status feedback

Message Name	ID	Cycle (ms)	Message Length (Byte)
bms_flag_Infor	0x18C4E2EF	100	8

Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Battery SOC	Intel	0	0	8	Unsigned	1	0	%	1%/bit;
Single cell overvoltage protection	Intel	1	8	1	Unsigned	1	0		0 = off 1 = on
Single cell undervoltage	Intel	1	9	1	Unsigned	1	0		0 = off 1 = on

protection									
Overvoltage protection for the entire battery pack	Intel	1	10	1	Unsigned	1	0		0 = off 1 = on
Undervoltage protection for the entire battery pack	Intel	1	11	1	Unsigned	1	0		0 = off 1 = on
Charging over temperature protection	Intel	1	12	1	Unsigned	1	0		0 = off 1 = on
Charging under temperature protection	Intel	1	13	1	Unsigned	1	0		0 = off 1 = on
Discharging over temperature protection	Intel	1	14	1	Unsigned	1	0		0 = off 1 = on
Discharging under temperature	Intel	1	15	1	Unsigned	1	0		0 = off 1 = on
Charging over current protection	Intel	2	16	1	Unsigned	1	0		0 = off 1 = on
Discharging over current protection	Intel	2	17	1	Unsigned	1	0		0 = off 1 = on
Short circuit protection	Intel	2	18	1	Unsigned	1	0		0 = off 1 = on
Fore-end IC error detection	Intel	2	19	1	Unsigned	1	0		0 = off 1 = on
Software locks up MOS	Intel	2	20	1	Unsigned	1	0		0 = off 1 = on
Charging status	Intel	2	21	2	Unsigned	1	0		0 = not charging 1 = manual charging adapter 2 = front charging station 3 = rear charging station
The current highest temperature of battery	Intel	3	28	12	signed	0.1	0	℃	0.1℃/bit;
The current lowest temperature of battery	Intel	4	40	12	signed	0.1	0	℃	0.1℃/bit;
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6

Driving motor torque status feedback									
Message Name			ID			Cycle (ms)		Message Length (Byte)	
Drive_fb_MCUEcoder			0x18C4DCEF			10		8	
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Left driving motor torque feedback	Intel	0	0	16	signed	0.01	0		
Right driving motor torque feedback	Intel	2	16	16	signed	0.01	0		
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6

Ultrasonic radar 1 - feedback frame(reserved)									
Message Name			ID			Cycle (ms)		Message Length (Byte)	
ultrasonic_1_fb			0x18C4E8EF			100		8	
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
No.1 ultrasonic radar probe’s distance	Intel	0	0	12	Unsigned	1	0	mm	This frame signal exists when

No.2 ultrasonic radar probe's distance	Intel	1	12	12	Unsigned	1	0	mm	ultrasonic radar is connected; otherwise, it remains reserved.
No.3 ultrasonic radar probe's distance	Intel	3	24	12	Unsigned	1	0	mm	
No.4 ultrasonic radar probe's distance	Intel	4	36	12	Unsigned	1	0	mm	
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6

Ultrasonic radar 2 - feedback frame(reserved)									
Message Name			**ID**				**Cycle (ms)**		**Message Length (Byte)**
ultrasonic_2_fb			0x18C4E9EF				100		8
Signal Description	**Format**	**Starting Byte**	**Starting Bit**	**Signal Length**	**Data Type**	**Precision**	**Offset**	**Unit**	**Signal Value Description**
No.5 ultrasonic radar probe's distance	Intel	0	0	12	Unsigned	1	0	mm	This frame signal exists when ultrasonic radar is connected; otherwise, it remains reserved.
No.6 ultrasonic radar probe's distance	Intel	1	12	12	Unsigned	1	0	mm	
No.7 ultrasonic radar probe's distance	Intel	3	24	12	Unsigned	1	0	mm	
No.8 ultrasonic radar probe's distance	Intel	4	36	12	Unsigned	1	0	mm	
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR

									Byte3 XOR Byte4 XOR Byte5 XOR Byte6

Vehicle fault status feedback									
Message Name			**ID**				**Cycle (ms)**		**Message Length (Byte)**
Veh_fb_Diag			0x18C4EAEF				10		8
Signal Description	**Format**	**Starting Byte**	**Starting Bit**	**Signal Length**	**Data Type**	**Precision**	**Offset**	**Unit**	**Signal Value Description**
Fault level	Intel	0	0	4	Unsigned	1	0		0: no fault 1: level 1 fault 2: level 2 fault 3: level 3 fault Other alarm invalid
CAN communication error under Auto control	Intel	0	4	1	Unsigned	1	0		0 = normal 1 = failed
CAN communication error under Auto IO control	Intel	0	5	1	Unsigned	1	0		0 = normal 1 = failed
EPS error	Intel	1	8	12	Unsigned	1	0		0 = normal error status as per EPSMCU_Note indicate
Left driving motor controller error	Intel	4	32	6	Unsigned	1	0		0 = normal error status as per DrvMCU_Note indicate
Right driving motor controller error	Intel	4	38	6	Unsigned	1	0		0 = normal error status as per DrvMCU_Note indicate
BMS CAN communication offline error	Intel	5	44	1	Unsigned	1	0		0 = normal 1 = failed
E-stop error	Intel	5	45	1	Unsigned	1	0		0 = open 1 = close
Remote controller power off alarm	Intel	5	46	1	Unsigned	1	0		0 = normal 1 = failed
Remote controller receiver offline error	Intel	5	47	1	Unsigned	1	0		0 = normal 1 = failed
Auxiliary error reserved	Intel	5	48	4	Unsigned	1	0		

Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6

Note: EPSMCU_Note explaination

Signal ID	Name	Description	Level
0x00		EPS No Fault	0
0x01	DiagEPS_InvalidSt	EPS Driver Status Error	3
0x02	DiagEPS_MotorFailed	EPS Motor Abnormal	3
0x03	DiagEPS_SensorlessEstimatorFailed	EPS Sensorless Abnormal	3
0x04	DiagEPS_EncoderFailed	EPS Encoder Abnormal	3
0x05	DiagEPS_ControllerFailed	EPS Controller Abnormal	3
0x06	DiagEPS_MinEndStopPressed	EPS Low Limit Triggered	1
0x07	DiagEPS_MaxEndStopPressed	EPS High Limit Triggered	1
0x08	DiagEPS_Estop	EPS Emergency Stop	3
0x09	DiagEPS_HomingWithoutEndStop	EPS Homing Without Limit Switch	1
0x0A	DiagEPS_UnKnownPosition	EPS No Position Information	1
0x0B	DiagEPS_PhaseResistanceOutOfRange	EPS Phase Resistance Out of Normal Range	3
0x0C	DiagEPS_PhaseInductanceOutOfRange	EPS Phase Inductance Out of Normal Range	3
0x0D	DiagEPS_ControlDeadlineMissed	EPS FOC Frequency Too High	3
0x0E	DiagEPS_ModulationMagnitude	EPS SVM Modulation Abnormal	3
0x0F	DiagEPS_CurrentSenseSaturation	EPS Phase Current Saturation	3
0x10	DiagEPS_CurrentLimitVolation	EPS Motor Current Overload	2
0x11	DiagEPS_MotorThermistorOverTemp	EPS Motor Overheating	2
0x12	DiagEPS_FetThermistorOverTemp	EPS Driver Overheating	2
0x13	DiagEPS_timerUpdateMissed	EPS FOC Processing Delay	2
0x14	DiagEPS_CurrentMeasurementUnavailable	EPS Phase Current Sampling Loss	2
0x15	DiagEPS_ControllerFailed	EPS Control Abnormal	3
0x16	DiagEPS_IBUSOutOfRange	EPS Bus Current Overlimit	2
0x17	DiagEPS_BrakeResistorDisarmed	EPS Braking Resistor Drive Abnormal	3
0x18	DiagEPS_SystemFailed	EPS System-Level Abnormal	3

0x19	DiagEPS_BadTiming	EPS Phase Current Sampling Delay	3
0x1A	DiagEPS_UnKnownPosition	EPS Motor Position Unknown	3
0x1B	DiagEPS_UnknownSpeed	EPS Motor Speed Unknown	2
0x1C	DiagEPS_UnknownTorque	EPS Torque Unknown	2
0x1D	DiagEPS_UnknownCurrentMeasurement	EPS Torque Control Unknown	2
0x1E	DiagEPS_UnknownCurrentMeasurement	EPS Current Sampling Value Unknown	2
0x1F	DiagEPS_UnknownVbusVoltage	EPS Voltage Sampling Value Unknown	2
0x20	DiagEPS_UnknownVoltageCtrl	EPS Voltage Control Unknown	2
0x21	DiagEPS_UnknownCurrentGain	EPS Current Loop Gain Unknown	2
0x22	DiagEPS_CtrlInitializing	EPS Controller Initialization Abnormal	2
0x23	DiagEPS_UnbalancedPhase	EPS Three-Phase Imbalance	2
0x24	DiagEPS_UnstabelGain	EPS Encoder Bandwidth Too High	3
0x25	DiagEPS_CPRPolepaireMismatch	EPS CPR and Pole Pairs Mismatch	3
0x26	DiagEPS_NoResponse	EPS Encoder No Response	3
0x27	DiagEPS_SecEncComFail	EPS Secondary Encoder Communication Error	3
0x28	DiagEPS_OverSpeed	EPS Speed Too High	2
0x29	DiagEPS_InvalidInputMode	EPS Control Input Mode Incorrect	2
0x2A	DiagEPS_UnstableGain	EPS PLL Gain Unstable	2
0x2B	DiagEPS_InvalidEstimate	EPS Position/Speed Unstable	2
0x2C	DiagEPS_SpinoutDetected	EPS Encoder Calibration Incorrect or Magnet Unstable	3
0x2D	DiagEPS_DcBusUnderVoltage	EPS Power Supply Undervoltage	2
0x2E	DiagEPS_DcBusOverVoltage	EPS Power Supply Overvoltage	2
0x2F	DiagEPS_DcBusOverRegenCurrrent	EPS Power Reverse (Charging) Current Overload	2
0x30	DiagEPS_DcBusOverCurrent	EPS Power Reverse (Discharging) Current Overload	2
0x31	DiagEPS_DisOnline	EPS Driver Offline	3

Note: DrvMCU_Note explaination

Signal ID	Name	Description	Level
0x00		Drive No Fault	0
0x01	diagDrvMCU_InternalError	Internal Error	3
0x02	diagDrvMCU_EncoderABZError	Encoder ABZ Signal Error	3
0x03	diagDrvMCU_HallUVWError	Encoder UVW Signal Error	3
0x04	diagDrvMCU_EncoderCntError	Encoder Counting Error	3
0x05	diagDrvMCU_MCUOT	Driver Overheating	2
0x06	diagDrvMCU_MCUOV	Driver Bus Overvoltage	2
0x07	diagDrvMCU_MCUUV	Driver Bus Undervoltage	2
0x08	diagDrvMCU_OutputShorCircuit	Driver Output Short Circuit	3
0x09	diagDrvMCU_BrakeResistanceOT	Driver Braking Resistor	2

		Overtemperature Warning	
0x0A	diagDrvMCU_FollowingError	Actual Following Error Exceeds Allowable Value	2
0x0B	diagDrvMCU_McuMotorOverLoad	I^2T Fault (Driver or Motor Overload)	2
0x0C	diagDrvMCU_SpdFollowingError	Speed Following Error Exceeds Allowable Value	2
0x0D	diagDrvMCU_MotorOT	Motor Overtemperature Warning	2
0x0E	diagDrvMCU_EncoderComError	Motor Homing Error (Communication-Based Encoder)	3
0x0F	diagDrvMCU_VcuDisOnline	Driver Detects VCU Communication Loss Warning	3
0x10	diagDrvMCU_McuDisOnline	VCU Detects Driver Offline	3

4.3.2. CAN cable connection

The CAN wires of FR-mini have all been soldered and labeled for easy connection by users according to the markings, as shown in Figure 4-1

Figure 4-1 CAN cable

4.3.3. VCU protocol instruction

1. Precautions During Testing:

1.1 During transmission, ensure the AliveCounter changes continuously in a cyclic manner.

1.2 Pay special attention to the fact that the AliveCounter occupies bits 52 to 55 (four bits).

1.3 The checksum in BYTE[7] is the XOR of the first 7 bytes:
Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6

1.4 The following routine is a simplified control command when sending via USB CAN. When controlling the vehicle, ensure commands are sent according to the communication protocol.

1.5 During testing, switch the remote controller to autonomous mode or turn it off.

1.6 When testing with a CAN adapter connected to a PC, since the vehicle may move, lift the vehicle during testing. Only lower it after ensuring stability in the test program.

1.7 During ground testing, since the remote controller has the highest priority, it is best to keep it on for easy switching to remote control mode if needed.

2. Control Command Description (ctrl_cmd)

The vehicle control command must be sent together with the corresponding command, heartbeat signal, and checksum.

(1) Target Gear Request (ctrl_cmd_gear)

The **ctrl_cmd_gear** command represents the target gear signal, with a physical value range of 01 to 04. The default gear position is 01 (Park - P).

03: Neutral (N)
02: Reverse (R)
04: Drive (D)
01: Park (P)

Example:
If the target gear request is Drive (D), the value is 04 (0x04).

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D2D0	0x04	0x00	0x00	0x00	0x00	0x00	0x10	0x14
0x18C4D2D0	0x04	0x00	0x00	0x00	0x00	0x00	0x20	0x24
0x18C4D2D0	0x04	0x00	0x00	0x00	0x00	0x00	0x30	0x34

Note: The above three frames should be cyclically transmitted at 10ms intervals to enable gear shifting to D (Drive) mode.

Return:

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D2EF	0x04	0x00	0x00	0x00	0x00	0x00	0x00	0x04

Note: The checksum and AliveCounter change cyclically.

(2) Target Speed Request (ctrl_cmd_velocity)

The **ctrl_cmd_velocity** command specifies the target speed value for vehicle propulsion, with a CAN communication physical range of 0 to 65.535 m/s (maximum speed of 6.6 m/s for a 10:1 speed ratio and 420mm wheel diameter). The target speed is determined by the speed resolution (0.001 m/s/bit).

Target vehicle speed = 0.001 * bus signal.

The vehicle's forward/reverse motion is coordinated with the gear position.

Speed feedback is provided in three modes:

1) **Current speed feedback:** Always returns a positive value.

2) **Left/right wheel speed feedback:** Indicates the respective speeds of the left and right wheels (positive for forward, negative for reverse).

3) **Left/right wheel pulse count feedback:** Pulse count increments during forward motion and decrements during reverse.

Example:

For a forward speed request of 5 m/s, the bus signal value is 5000 (0x1388).

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D2D0	0x84	0x38	0x01	0x00	0x00	0x00	0x00	0xBD
0x18C4D2D0	0x84	0x38	0x01	0x00	0x00	0x00	0x10	0xAD
0x18C4D2D0	0x84	0x38	0x01	0x00	0x00	0x00	0x20	0x9D

Note: The above three frames should be transmitted cyclically at 10ms intervals to maintain the vehicle's forward motion at 5m/s.

Return:

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D2EF	0x84	0x38	0x01	0x00	0x00	0x00	0x00	0xBD

Note: The checksum and AliveCounter change cyclically. Due to automatic speed adjustment during operation, the actual feedback speed may not be exactly 5m/s.

Left wheel speed and pulse feedback ID: 0x18C4D7EF
Right wheel speed and pulse feedback ID: 0x18C4D8EF

(3) Target Steering Angle (ctrl_cmd_steering)

The ctrl_cmd_steering command specifies the target steering angle request, with a CAN communication physical range of -40.96° to +40.95°. The vehicle's internal software limits the steering angle to -32° to +32°.

Left turn: Positive value
Right turn: Negative value

The target steering angle is determined by a resolution of 0.01°/bit:
Target steering angle = Bus signal × 0.01

Example:
For a target steering angle of -25°, the bus signal value is -2500 (0xF63C).

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D2D0	0x00	0x00	0XC0	0x63	0x0F	0x00	0x00	0xAC
0x18C4D2D0	0x00	0x00	0XC0	0x63	0x0F	0x00	0x10	0xBC
0x18C4D2D0	0x00	0x00	0XC0	0x63	0x0F	0x00	0x20	0x8C

Note: The above three frames should be transmitted cyclically at 10ms intervals to set the steering angle request to -25 degrees.

Return:

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D2EF	0x00	0x00	0XC0	0x63	0x0F	0x00	0x00	0xAC

Note: The checksum and AliveCounter change cyclically.

(4)Auxiliary Control Command Description

Taking the clearance lamp enable as an example (other auxiliary controls follow the same principle as clearance lamp enable control). The IO port enable control requires simultaneous transmission of the enable flag, heartbeat signal, and checksum. (If IO control is disabled, all lighting controls will be managed by the VCU.)

Example:

io_cmd_clearance_lamp - Clearance lamp enable control

0x01: Enable

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D7D0	0x01	0x20	0X00	0x00	0x00	0x00	0x00	0x21
0x18C4D7D0	0x01	0x20	0X00	0x00	0x00	0x00	0x10	0x31
0x18C4D7D0	0x01	0x20	0X00	0x00	0x00	0x00	0x20	0x01

Note: The above three frames should be transmitted cyclically at 50ms intervals to activate the high beam.

Return:

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4DAEF	0x01	0x20	0X00	0x00	0x00	0x00	0x00	0x21

Note: The checksum and AliveCounter change cyclically.

The auxiliary enable control supports clearance lamp control and left/right turn signal control. The brake lights are not controlled by CAN signals but are entirely managed by the VCU, which provides feedback on their enable status.

5. Precautions

This section contains some important considerations for the use, storage, and development of the FR-mini.

5.1. Battery Usage Precautions

▲ The FR-mini product may not be fully charged when shipped. The remaining battery level can be checked via the FR-mini remote control's vehicle chassis display or through the CAN bus communication interface. Charging is complete when the charger's green indicator light turns on.

▲ Do not wait until the battery is completely depleted before recharging. Recharge promptly when the remaining battery level drops below 10%.

▲ The operating temperature range for battery discharge is -20°C to 50°C. The battery functions normally within this range; exceeding it may trigger battery protection.

▲ Charge the battery at temperatures above 0°C. Below 0°C, the battery may enter protection mode and refuse to charge.

▲ Avoid excessive discharge during use, as it may damage the battery.

▲ Prevent excessive impact on the battery, as impacts beyond specifications may cause leakage, overheating, smoking, fire, or explosion.

▲ If the battery shows obvious abnormalities, stop using it immediately!

5.2. Charging Precautions

▲ Only use the dedicated charger provided with the battery. Do not use non-standard batteries, power supplies, or chargers.

▲ Charge only in environments between 10°C and 45°C. Charging outside this range may cause battery leakage, overheating, severe damage, or deterioration in performance and lifespan.

▲ If the charger or battery behaves abnormally or is damaged during charging, unplug the charger's input and output power cables immediately.

▲ If charging cannot be completed within the specified time, stop the charging process. Otherwise, the battery may overheat, emit smoke, catch fire, or explode.

▲ Do not charge the vehicle battery during thunderstorms.

▲ Do not charge the vehicle battery in damp, rainy, or submerged locations.

▲ Do not charge the vehicle battery near heat sources or in direct sunlight where temperatures are excessively high.

▲ Charge in a well-ventilated, dust-free area.

▲ Do not block the charger's air vents during charging; maintain at least 10cm of clearance.

5.3. Usage Environment Precautions

▲ The FR-mini operates between -20°C and 50°C. Do not use it in temperatures below -20°C or above 50°C.

▲ The optimal storage temperature for the FR-mini is 0°C to 25°C.

▲ Do not store or use in environments with corrosive, flammable, or explosive gases.

▲ Keep away from heat sources and open flames during use and storage.

▲ Unless it is a specially customized version (with IP protection rating), the FR-mini has limited waterproofing. Avoid using it in deep water.

5.4. Remote Control Operation Precautions

▲ When debugging with the remote control, ensure it is powered on and that the vehicle can receive commands. If the remote signal is weak, the remote will alarm—move closer to the vehicle.

▲ Before powering on, ensure all DIP switches are in the uppermost position, the emergency stop switch is released, and the throttle lever is at zero (chassis speed = 0).

▲ When using remote control, prioritize low-speed gear operation. Test medium or high speeds only after becoming familiar with the vehicle.

5.5. Electrical Expansion Precautions

▲ The top expansion power supply current must be used strictly according to the selected battery voltage and current specifications. Do not overload.

▲ When plugging or unplugging the top expansion power interface, turn off the vehicle switch to avoid short circuits.

▲ If the system detects that the battery voltage is below the safe threshold, it will activate protection. If external expansion devices involve critical data storage without automatic power-loss protection, recharge promptly.

5.6. Other Precautions

▲ Do not drop or invert the device during transportation or setup.

▲ Do not disassemble the device without professional authorization.

▲ If the remote control terminal is unused for an extended period, remove its batteries.

▲ Replace tires promptly if the tread pattern shows significant wear.

6. Common Q&A

Q: The FR-mini starts normally, but the vehicle does not move when controlled by the remote. What should I do?

A: First, check if the rear emergency stop switch is released. Then verify whether the SWA lever is in remote control mode, whether the VRB knob is unlocked, and whether the SWB gear lever matches the control command.

Q: What should I do if the FR-mini remote runs out of power and the vehicle stops?

A: Replace the remote control batteries immediately. Communication will resume once the remote is powered on again.

Q: Can the chassis shell be removed for vehicle body modifications?

A: This chassis uses a non-load-bearing body design, meaning the shell does not support internal components. Removing the shell has minimal impact, but it is not recommended. If necessary, have it done by professionals. The company is not responsible for damages caused by unauthorized disassembly.

Q: Neither LED1 nor LED2 on the charger lights up.

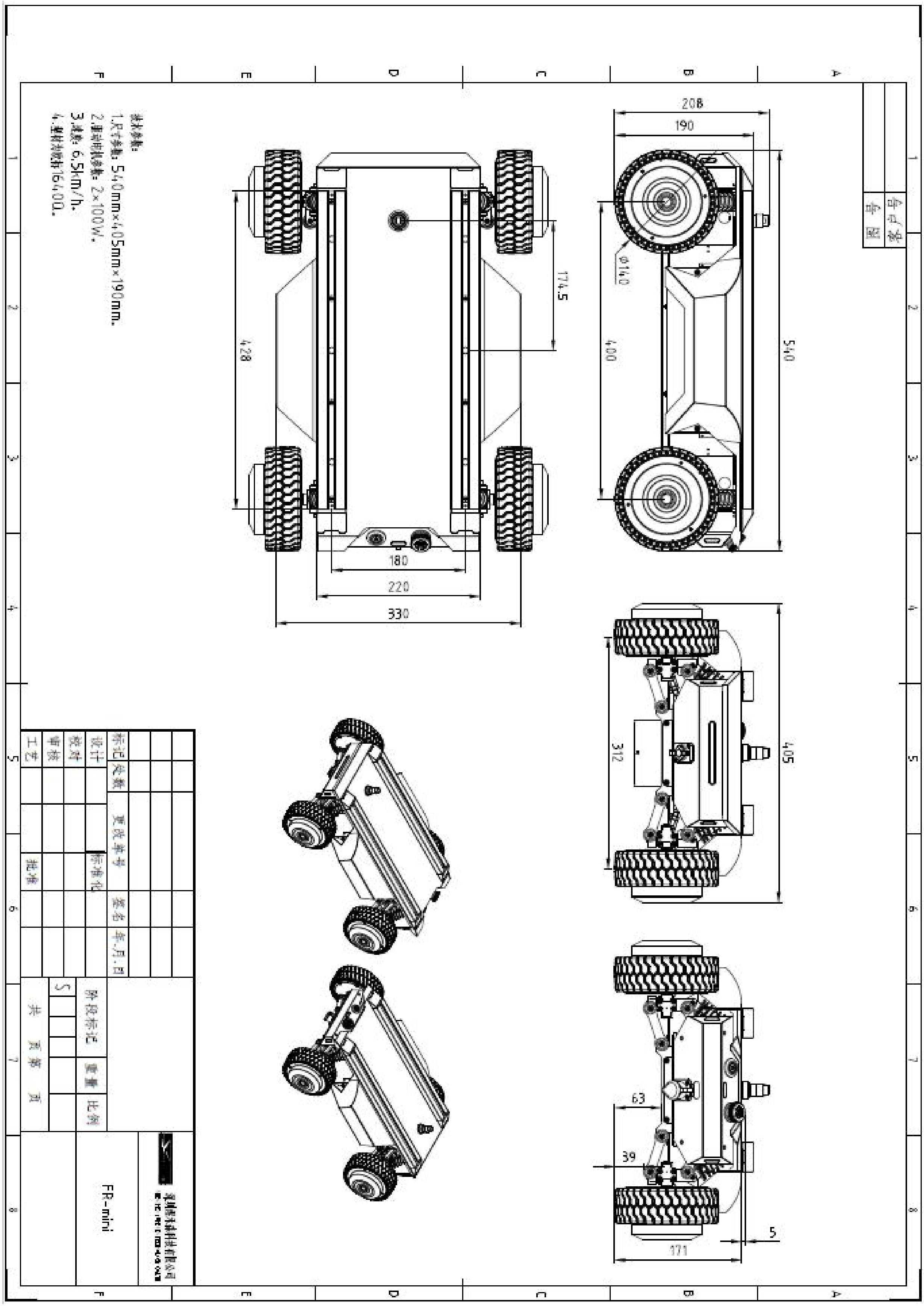
A: First, check if the charger's input cable is securely connected. Then verify whether there is AC power input.

- Is the battery unused for a long time, over-discharged, or damaged?
- Replug the input and output cables with an interval of more than 10 seconds to check if the charger is in protection mode.

Q: How to turn off the remote control while the receiver remains powered?

A: Remove and reinsert the remote control batteries to shut down the remote while keeping the receiver powered.

7. Specification



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